

AMOGH SOMISETTY

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SUMMARY

Electrical Engineering student focused on digital design and design verification. Designed a multi-cycle **RISC-V (RV32I) CPU in SystemVerilog** and synthesized, placed-and-routed, and verified it on a Basys-3 FPGA through the Xilinx Vivado (Tcl-driven) flow. Strong RTL and FSM fundamentals paired with hands-on debug — resolved 6+ documented failure modes via systematic root-cause analysis. Automates EDA flows with Tcl, Python, and shell scripting.

EDUCATION

California Polytechnic State University, San Luis Obispo

B.S. Electrical Engineering

San Luis Obispo, CA

Expected June 2028 · Junior standing, Fall 2026

Relevant coursework: Computer Design & RISC-V Architecture (CPE 233), Digital Design (CPE 133), Electronics & Analog Circuits (EE 241), Circuit Analysis & Instrumentation (EE 143), Manufacturing Processes (IME 144), Materials Engineering (MATE 210/215)

TECHNICAL SKILLS

- **HDL & Digital Design:** SystemVerilog, RTL design, finite-state machines (Moore/Mealy), RISC-V RV32I ISA, datapath & control-unit design, combinational/sequential logic, timing analysis
- **Design Verification & Debug:** testbench simulation, waveform debugging, functional validation against ISA/spec, root-cause analysis, failure-mode documentation, test-plan mindset
- **EDA & Tools:** Xilinx Vivado (synthesis, place & route, timing closure), Basys-3 FPGA, LTspice, Autodesk Fusion 360
- **Scripting & Automation:** Tcl (Vivado synthesis/PnR flows), Bash/shell, Python, Git, GitHub Actions (CI)
- **Embedded & Analog:** C/C++ (Arduino/embedded), op-amp signal conditioning, ADC/DAC systems, PCB design (schematic → layout → reflow)

PROJECTS

OTTER Multi-Cycle RISC-V CPU — SystemVerilog · Vivado · Basys-3 FPGA (CPE 233)

- Designed a complete multi-cycle **RV32I** processor from gates up in SystemVerilog: PC → instruction memory → decode → 32×32 dual-read register file → 10-operation ALU → dual-port memory → writeback.
- Implemented a 2-state (FETCH/EXEC) FSM control unit and a branch-condition generator (br_eq, br_lt, br_ltu); built an immediate generator covering I/S/B/U/J instruction formats.
- Synthesized, placed-and-routed, and verified the design on a Basys-3 FPGA via the Vivado flow; debugged a branch-target miscalculation by gating PCWrite to the EXEC state and validating against the RISC-V ISA reference.

Digital Systems & FSM Design — Vivado · FPGA (CPE 133)

- Built combinational and sequential blocks — MUXes, decoders, D-flip-flops, shift registers, counters, clock dividers — and implemented FSMs using binary, one-hot, and gray state encodings, then synthesized and place-and-routed them onto FPGA hardware.

End-to-End Analog ↔ Digital Signal System + 4-bit DAC — PCB · LTspice (EE 143)

- Designed, simulated (LTspice), fabricated, and integrated a **4-bit binary-weighted DAC** (LM1458, 1206 SMD) on a 1×1" 2-layer PCB within a full audio → Arduino ADC → DAC → speaker pipeline.
- Achieved 62.5 mV/step resolution with measured-vs-theoretical error within ±1 LSB across all 16 codes; resolved analog loading with a unity-gain buffer.

9-Stage Electromechanical System — Embedded C++ · Analog Design (EE 241)

- Integrated a sequentially dependent 9-stage system (capacitive sensing, relay isolation, 580 V strobe, Schmitt-trigger detector, solenoid launcher, 555-timer metal detector, electromagnet release, tilt switch, LCD).
- Implemented frequency-domain metal detection (5.1% shift) with streak-based filtering; **diagnosed and resolved 6 distinct failure modes** across electrical, mechanical, and firmware domains via root-cause analysis.

EXPERIENCE

Natera — Engineering/Operations Intern

Pleasanton, CA · Summer 2025

- Mapped a validation-and-packaging workflow, identified bottlenecks, and removed redundant verification steps, cutting cycle time from ~20 min to ~10–12 min (observed estimate).
- Standardized the workflow and authored reference documentation adopted by the team for consistency.

CVS Pharmacy — Pharmacy Technician (CA state certified)

Dublin, CA · Jun 2023 – Jun 2024

- Maintained accuracy and regulatory compliance under time pressure in a high-volume retail pharmacy.

LEADERSHIP & ACTIVITIES

- **Poly-Engineering Consulting (Cal Poly):** designed and shipped an internal role-aware operating system (PEC Nexus) for project execution, scheduling, and permissions.
- Varsity wrestling (4 yrs) · Eagle Scout candidate (11 yrs, Scouting) · CPR/First Aid certified.